Prediction, Regularization, and Applied Measurement

Empirical Methods — Week 9

Teaching deck draft

Week 9 question

- When is prediction itself the research object?
- When is prediction an input into measurement or design?
- How do train/test logic, regularization, and calibration discipline the object?
- What can go wrong when a predicted variable enters an economics paper?

Machine learning can improve measurement and diagnostics, but it does not define the estimand or identify a causal effect.

Prediction versus causal design

Use	Main object	Evaluation question
Prediction as final object	risk, forecast, ranking, probability	Does it generalize and calibrate on the target population?
Prediction as measurement	score, label, exposure, proxy	Does it measure the economic construct needed downstream?
Prediction as nuisance input	flexible controls or nuisance functions	Does it help estimate the causal object without becoming the answer?

- Prediction accuracy is not identification.
- A good score can still be a bad measure if the target, timing, or population is wrong.
- The downstream design determines which prediction errors matter.

Train/test risk

$$R(f) = \mathbb{E}\left[L(Y, f(X))\right]$$

$$\widehat{f} = \arg\min_{f \in \mathcal{F}} \frac{1}{|\mathcal{I}_{train}|} \sum_{i \in \mathcal{I}_{train}} L(Y_i, f(X_i))$$

$$\widehat{R}_{test}(\widehat{f}) = \frac{1}{|\mathcal{I}_{test}|} \sum_{i \in \mathcal{I}_{test}} L(Y_i, \widehat{f}(X_i))$$

- Random splits answer a narrow generalization question.
- Time, firm, region, or occupation splits test transportability.
- The loss should match the economics cost of the mistake.

Regularization and tuning

$$\widehat{\beta}^{lasso} = \arg\min_{\beta} \frac{1}{n} \sum_{i=1}^{n} (Y_i - X_i'\beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

$$\widehat{\beta}^{ridge} = \arg\min_{\beta} \frac{1}{n} \sum_{i=1}^{n} (Y_i - X_i'\beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

$$\widehat{\beta}^{EN} = \arg\min_{\beta} \frac{1}{n} \sum_{i=1}^{n} (Y_i - X_i'\beta)^2 + \lambda \left[\alpha \sum_{j} |\beta_j| + (1-\alpha) \sum_{j} \beta_j^2 \right]$$

Cross-validation selects prediction rules

$$\widehat{\lambda} = \arg\min_{\lambda \in \Lambda} \frac{1}{K} \sum_{k=1}^{K} \frac{1}{|V_k|} \sum_{i \in V_k} L\left(Y_i, \widehat{f}_{-k,\lambda}(X_i)\right)$$

Penalty	Best when	Caveat
Lasso	sparse signal is plausible	selected variables can be unstable
Ridge	many related predictors matter	no sparse interpretation
Elastic net	grouped or collinear features	tuning has two moving parts

Cross-validation is a tuning device. It is not an exclusion restriction, a research design, or a policy-invariance argument.

Bias, variance, and calibration

$$\mathbb{E}\left[(Y - \widehat{f}(x))^2 \mid X = x\right] = \sigma^2 + \left(\mathbb{E}[\widehat{f}(x)] - f_0(x)\right)^2 + \mathrm{Var}(\widehat{f}(x))$$

$$\mathbb{E}[Y_i \mid \widehat{p}_i = s] = s, \qquad \widehat{B} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \widehat{p}_i)^2$$

$$\widehat{C}_i(t) = \mathbf{1}\{\widehat{p}_i \geq t\}$$

- Regularization accepts bias to reduce variance.
- Calibration matters when a score is used as probability or intensity.
- Thresholds encode policy or measurement tradeoffs.

Economics applications to measurement

Setting	Predicted object	Downstream use
Job postings	skills, tasks, technology labels	labor demand, task change, technology exposure
Raw job titles	occupation or task group	mobility, wage growth, exposure by occupation
Remote work	feasibility or intensity score	spatial sorting, firm policy, worker access
Risk environments	default, attrition, take-up	targeting, diagnostics, policy triage
Latent outcomes	imputed score or proxy	measurement when outcomes are missing or costly

The question is not "which algorithm is strongest?" It is whether the predicted object matches the economic construct.

Feature engineering and leakage

- Predictors must be available at the time and level of the intended prediction.
- Leakage can enter through post-outcome variables, post-treatment text, edited metadata, future codes, or labels embedded in features.
- Job-posting text, occupation codes, firm histories, and worker records all require a timing diagram.
- A model can pass a random holdout test while failing the actual research-use test.

$$\text{allowed feature} \quad \Longleftrightarrow \quad X_i \text{ observed before } Y_i \text{ and before the policy/design decision.}$$

Implementation pitfalls

Pitfall	Why it matters	Check
Class imbalance	accuracy hides rare-event failure	precision, recall, confusion matrix
Drift	language and behavior change	time-split validation
Transportability	model fails across firms or regions	group-split validation
Instability	selected features are sample-specific	fold-level selection stability
Subgroup error	measurement error is systematic	subgroup loss and calibration
Interpretability	measure is not auditable	target, features, labels, failure modes

Research Lab design

Reproduce

- synthetic job postings
- keyword baseline
- ridge, lasso, elastic net
- cross-validation and test loss

Diagnose

- calibration bins
- subgroup performance
- leakage audit
- feature stability

Transfer

- occupation-title data
- vocabulary coverage
- transportability loss
- downstream-use memo

Takeaways

- Start with the economic object, not the algorithm.
- Train/test discipline estimates generalization only for the population implied by the split.
- Regularization is useful because high-dimensional data are noisy and collinear.
- A predicted measure needs calibration, subgroup diagnostics, leakage checks, and transportability evidence.
- ML improves applied economics when it sharpens measurement while leaving identification explicit.